

Douglas J. Williams

AI-Assisted Solutions Builder · Technical Product Owner

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Open to fully remote roles

SUMMARY

Technical builder with 28+ years in mission-critical network operations who now delivers working software using AI-assisted, spec-driven development. My strength is ownership end-to-end: I define the vision and requirements for a tool, direct AI coding agents (Kiro) to build it in Python, then stay with the project through testing and debugging until it actually works and meets the need. At CPKC I drove a company-wide event-analysis platform that ingests billions of messages per day for proactive problem isolation. A deep operations background (SMS/MMS platforms, IMS Core, Kubernetes) means I understand the systems these tools serve and can tell when the output is right. Seeking a remote role building internal tools and applications with generative AI.

CORE SKILLS

AI-Assisted Development: Spec-driven development, requirements definition, directing AI coding agents (Kiro) to build in Python, iterative testing & debugging to a working result

Product Ownership: Translating business needs into a clear end-goal, scoping, seeing projects through to delivery, verifying output against real-world requirements

Tech I've shipped with (AI-directed): React, Supabase (Postgres), Python data ingestion, Vercel deployment, SQL & Oracle, HTML/CSS, Git/GitHub

Technical Foundation: Linux/command-line, log analysis at scale, reading and troubleshooting output

Systems Knowledge: Kubernetes / containerized platforms, virtualized messaging (SMS/MMS), IMS Core

Domain: Positive Train Control (PTC) & rail, telecom / network operations, oil & gas

Operations: 24/7 mission-critical support, incident response, vendor management, technical documentation (MOPs), mentoring

EXPERIENCE

Senior PTC Applications Engineer — CPKC (Canadian Pacific Kansas City)

2022 – Present · Remote

- Own the vision and delivery of internal tools supporting Positive Train Control (PTC) operations across the rail network — a safety-critical environment where the tool has to work.
- Drove development of a company-wide event-analysis platform that ingests billions of PTC messages per day, used across all of CPKC for detailed event analysis and proactive problem isolation, surfacing issues before they impact operations.
- Delivered Python database-integration tools connecting SQL and Oracle data sources into custom applications and reporting workflows.
- Work through AI-assisted, spec-driven development (Kiro): define requirements and the target result, direct the AI to build in Python, then test and debug iteratively until the tool behaves correctly and meets the operational need.

ATS Senior Network Engineer — T-Mobile

Aug 2020 – Aug 2022

- Product Lead for MCO (Message Controller) and Kubernetes-containerized MCO handling T-Mobile SMS and MMS traffic.
- Managed and implemented migration of Sprint traffic to a new Replica MCO environment, enabling contract expirations and consolidation onto a single network.

- Team lead across new design and implementation; drove proactive tooling and processes to identify and resolve issues before customer impact.
- Authored Methods of Procedure (MOPs) and trained teams on new products at rollout.

Network Translations Specialist / NTAC Network Engineer — Sprint

Jul 1998 – Aug 2020

- Implemented a new virtual messaging platform for SMS and MMS in coordination with Mavenir and Rackspace.
- Led migration of SMS traffic to the IMS Core via SIP with CDMA fallback.
- Migrated MMS storage and processing from Syniverse to an internal virtualized MMSC, eliminating outside managed-services costs.
- Saved Sprint \$1.2M over three years by renegotiating the SPAM-filtering vendor contract.
- Deployed Mavenir Spamshield, adding MMS SPAM filtering not previously offered.
- Provided 24/7/365 on-call support for all messaging elements during high-severity events.

SELECTED PROJECTS

PTC Event-Analysis Platform — built with Kiro (Python), SQL/Oracle (CPKC)

Company-wide tool that ingests billions of PTC messages per day for event analysis and proactive problem isolation. Owned the vision, requirements, and debugging through to a working, adopted result.

Database Integration Tools — built with Kiro (Python), SQL/Oracle (CPKC)

Utilities that pull operational data from source systems into custom apps and reports.

Oil & Gas Well-Tracker Application — React, Supabase, Python · built with Kiro (own business, J&T Cook Corporation)

Full-stack app to track oil & gas working-interest investments — investment vs. revenue, payout status, operator-statement ingestion, bank reconciliation, and tax reporting. Owned the vision and requirements, directed AI to build it, and debugged through to a live production deployment. Sanitized code: github.com/djwilli69/well-tracker-demo

BUSINESS OWNERSHIP

Owner — J&T Cook Corporation (Oil & Gas)

Own and operate an oil & gas business, including building its web presence with AI-assisted development. Brings a product owner's mindset: I understand the business goal behind the software, not just the task.

EDUCATION

Bachelor of Science, Business Management — Friends University, Wichita, KS